Stage 1
AI as Conversational Assistant

Stage 2
AI as Reactive Operator

Stage 3
AI as Adaptive Coordinator

Stage 4
AI as Self-Governing Production System

Nascent Diffusion

Growing Industries

Widespread Industries

Deep workflow integration· dual-mechanism productivity · governed autonomy

Information Tech

Applications

- **Auto Coding Systems**: Instruction-following in multi-round programming
- **Iterative Debugging**: test-guided refinement during coding...
- **Repository-Level Engineering**

Expansion Mechanism

- **Workflow Orchestration:** Coordinating multi-step software workflows
- **Role Specialization:** Distributing work across specialized agents
- **Environment-Driven Learning:** Real repository environments to strengthen autonomous performance

Finance

Applications

- **Investment and Trading Systems**: autonomous trade execution...
- **Banking Operations**: conversational execution of financial services...
- **Insurance Processing**

Expansion Mechanism

- **Autonomous Strategy Optimization:** Iterative refinement through reflection
- **Cross-Source Data Fusion:** Integration of transactions with unstructured signals
- **Multi-Agent Institutional Simulation**
- **Privacy-Preserving Collaboration:** cross-institution modeling

Medical

Applications

- **Clinical Systems**: Documentation automation...
- **Pharma R&D**: Molecular generation...
- **Public Health Services:** Continuous consultation...

Expansion Mechanism

- **Scalable Analysis:** Large-scale parallel processing
- **Cross-Modal Integration:** Joint reasoning across clinical data
- **Continuous Service**
- **Multi-Agent Orchestration:** Coordinated workflows across multistep